

Deepfake Generation Method Classification Using Artifact-Oriented Features

Yuval Felendler, Chen Frydman, Amit Avraham, and Mustafa Hijazi

Course Project: Offensive AI, Course Number: 237.2.6905, Course Instructor: Dr. Yisroel Mirsky,

Submit Date: 21/01/2026, Group number 3, Ben-Gurion University of the Negev

yuvalfel@post.bgu.ac.il, chenfryd@post.bgu.ac.il, avramit@post.bgu.ac.il, hijazimu@post.bgu.ac.il

KEYWORDS

Deepfake attribution, Deepfake method classification, Frequency-domain analysis, artifact-based detection, FaceForensics++, Multi-class deepfake detection

1 INTRODUCTION

The rapid advancement of face manipulation technologies has produced deepfake videos of near-photorealistic quality [11], posing challenges for both human observers and automated detection systems [19]. Most existing research addresses binary classification—determining whether content is authentic or manipulated. However, this formulation provides limited forensic value. Identifying which manipulation technique created a given video offers substantially greater utility [3]: it enables attribution to specific tools, characterizes adversarial capabilities, and supports targeted defense strategies adapted to particular generation methods.

Current deepfake techniques follow distinct technical paradigms. Autoencoder-based approaches like DeepFakes perform identity replacement by learning separate latent representations for source and target faces. Graphics-based methods such as Face2Face [22] transfer expressions using 3D facial tracking and texture mapping. Neural rendering systems like NeuralTextures [21] generate photorealistic appearances through learned texture models conditioned on geometric cues. Each paradigm leaves characteristic traces: autoencoder methods produce boundary inconsistencies [14] and spectral anomalies from upsampling [8]; tracking-based systems exhibit geometric instability during motion [17]; neural renderers fail to maintain temporal coherence across frames [26]. Yet two questions persist: which artifacts reliably separate specific method pairs, and can any single feature set achieve robust multi-class discrimination?

We investigate deepfake method classification using interpretable features designed to exploit known weaknesses

in generation pipelines. Instead of training end-to-end networks that may overfit to dataset biases, we engineer features targeting documented failure modes. We extract three complementary signals: frequency-domain statistics that expose upsampling artifacts [8], geometric stability measures that detect tracking failures, and optical flow residuals that quantify temporal discontinuities [1]. Through systematic pairwise and multi-class evaluation, we characterize both classification performance and diagnostic patterns—revealing which methods share overlapping artifacts and which remain separable. Our results show that successful attribution requires matching feature strategies to underlying generation mechanisms; no universal representation proves effective across all method pairs.

2 RELATED WORK

2.1 Deepfake Detection Datasets

The development of deepfake detection methods has been facilitated by large-scale benchmark datasets. FaceForensics++ [19] remains the most widely adopted benchmark, providing 1,000 original videos and 4,000 manipulated versions across four methods: DeepFakes, Face2Face, FaceSwap, and NeuralTextures. The dataset offers three compression levels to evaluate robustness under realistic conditions. CelebDF [15] addresses quality concerns in earlier datasets by providing 5,639 high-fidelity deepfake videos of celebrities, emphasizing visual realism over artifact prominence. The DeepFake Detection Challenge (DFDC) [7] dataset contains over 100,000 videos with diverse demographics and manipulation methods. DeeperForensics-1.0 [12] introduces controlled variation in environmental factors such as lighting and occlusion to test generalization. While these datasets enable supervised learning for binary detection, few works exploit their multi-method structure for attribution tasks.

2.2 Frequency-Domain Detection Methods

Frequency analysis has emerged as a powerful tool for exposing synthesis artifacts. Durall et al. [8] demonstrated that GAN (generative adversarial network)-based generators fail to reproduce natural spectral distributions due to upsampling operations, leaving characteristic fingerprints in the frequency domain. Frank et al. [9] leveraged these

findings by training classifiers on frequency representations, achieving strong generalization across unseen manipulation methods. Qian et al. [18] proposed mining frequency-aware clues through cross-band correlation analysis, showing that high-frequency components carry discriminative forgery traces. Luo et al. [16] extended this work by designing high-frequency feature extractors that generalize across different GAN architectures. More recently, Tan et al. [20] introduced frequency space learning to improve detector robustness. These works focus primarily on binary classification; we extend frequency analysis to method attribution by designing radial Fast Fourier transform (FFT) features that capture method-specific spectral signatures.

2.3 Temporal Coherence Analysis

Deepfake videos often exhibit temporal inconsistencies that static frame analysis cannot detect. Amerini et al. [1] pioneered the use of optical flow for deepfake detection, showing that inter-frame motion patterns differ between authentic and manipulated sequences. Caldelli et al. [5] extended this approach to cross-forgery scenarios, demonstrating that optical flow features generalize better than spatial features when tested on unseen manipulation methods. Zheng et al. [26] proposed modeling temporal coherence through long-range dependencies, capturing subtle frame-to-frame inconsistencies. Amin et al. [2] introduced temporal facial patterns that track color variations across forgery-sensitive regions. Our work employs optical flow but focuses on warping residuals rather than flow magnitude, specifically targeting temporal artifacts introduced by frame-by-frame synthesis.

2.4 Generation Methods and Their Artifacts

Understanding deepfake generation techniques is essential for designing targeted detection features. Face2Face [22] performs real-time expression transfer using 3D face tracking and texture mapping, introducing geometric instabilities under rapid motion. NeuralTextures [21] synthesizes photorealistic appearances through learned neural texture representations conditioned on geometric proxies, producing temporal flicker artifacts. DeepFakes, based on autoencoder architectures, performs identity replacement by learning disentangled latent representations, leaving boundary inconsistencies and spectral anomalies. FaceSwap employs classical 3D face alignment for identity transfer, exhibiting lighting inconsistencies. While prior work [14, 17] has identified these artifacts for binary detection, we systematically evaluate which artifacts enable discrimination between specific method pairs.

2.5 Artifact Attribution and Fingerprinting

Recent work has explored attributing synthetic images to specific generators. Yu et al. [24] demonstrated that GANs leave characteristic fingerprints that enable model-level attribution, showing that different architectures produce distinct artifact patterns. However, their focus remains on image generation rather than face manipulation. To our knowledge, few studies address multi-class deepfake method attribution using interpretable features. Most detection systems [6, 10] employ end-to-end deep networks trained for binary classification, offering limited insight into which artifacts drive predictions. Our approach differs by engineering features that explicitly target documented weaknesses in generation pipelines, enabling both classification and diagnostic analysis of method separability.

2.6 Detection Architectures and Evaluation

State-of-the-art deepfake detectors predominantly employ deep convolutional architectures. Xception [6] has become a standard baseline due to its depthwise separable convolutions that capture fine-grained manipulation traces. ResNet [10] variants are commonly used for feature extraction, particularly when combined with attention mechanisms. Recent work [23] has explored freezing spatial and temporal networks alternately to improve efficiency. Comprehensive reviews [11] categorize detection methods by feature type, network architecture, and evaluation protocol. However, most studies evaluate performance through binary metrics on held-out test sets, providing limited understanding of which methods share overlapping artifacts. We employ Random Forests [4] with stratified cross-validation [13] to enable feature importance analysis while maintaining robust performance estimates.

3 METHOD

3.1 Dataset

This study employs the FaceForensics++ dataset with c23 compression, which contains high-quality YouTube videos and their corresponding manipulated versions generated through multiple facial manipulation techniques. Unlike traditional deepfake detection tasks, this work focuses exclusively on manipulated videos, aiming to predict the specific manipulation method used rather than distinguishing authentic from fake content.

The dataset includes four manipulation pipelines:

DeepFakes: An autoencoder-based identity swap method employing separate encoder-decoder pairs per identity. The face region is replaced and blended into the target frame, often producing boundary blending errors and texture smoothing artifacts.

Face2Face: An expression reenactment technique that preserves target identity while transferring facial expressions via 3D face tracking. Characteristic artifacts include geometric jitter and instability around the mouth and eyes.

FaceSwap: A classical approach utilizing 3D face alignment and texture transfer for identity replacement with relatively stable geometry. Common artifacts include identity mismatch and lighting inconsistencies.

NeuralTextures: A neural rendering-based method that learns texture representations conditioned on pose to produce realistic individual frames. Typical artifacts include temporal flicker and high-frequency inconsistencies.

From each method, 120 manipulated videos are selected, each derived from a unique source video, yielding a balanced dataset of 480 videos (4 classes $\times$ 120 videos). These methods differ fundamentally in their synthesis targets—identity, expression, or texture—producing distinct artifact patterns suitable for comparative analysis.

3.2 Preprocessing and Frame Sampling

To obtain temporally representative inputs, 24 frames are uniformly sampled along each video’s duration, maintaining comparable feature dimensionality across samples while providing temporal coverage and reducing sensitivity to transient artifacts.

For each sampled frame, MTCNN (Multi-task Cascaded Convolutional Networks) localizes the primary face. The most confident detection is selected, and its bounding box is expanded by 20 pixels to include facial boundaries. The face region is then cropped and resized to 160×160 pixels in RGB format. Frames without successful face detection are discarded, and videos with fewer than 18 valid detections are excluded to ensure reliable facial alignment and temporal statistics computation.

3.3 Artifact-Oriented Feature Extraction

This method employs interpretable, artifact-driven features targeting known vulnerabilities in generation pipelines. All features are computed per video by aggregating statistics across sampled face crops.

3.3.1 Frequency-Domain Features (FFT). Neural generators employ convolutional upsampling, leaving characteristic traces in the frequency domain. For each cropped face, the grayscale image undergoes 2D Fast Fourier Transform (FFT) computation. The zero-frequency component is centered, and the magnitude spectrum is obtained on a logarithmic scale. Spectral energy is aggregated into three radial bands based on normalized radius: low frequency ($r < 0.2$), mid frequency ($0.2 \leq r < 0.5$), and high frequency ($r \geq 0.5$). For each band, mean and standard deviation across all frames

are calculated, yielding a 6-dimensional frequency feature vector per video.

3.3.2 Geometric Jitter Features. Reenactment and rendering methods often fail to maintain consistent facial geometry over time. To capture this instability, Canny edge detection (thresholds: 50 and 150) is applied to each face crop. Edge pixel coordinates are extracted, and the facial edge structure centroid is computed for each frame (excluding frames with fewer than 100 edge pixels). The centroid trajectory is tracked across the video, and eight temporal displacement statistics are calculated: mean, standard deviation, maximum, 90th percentile, 95th percentile, mean of second-order differences, standard deviation of second-order differences, and variance. This 8-dimensional descriptor serves as a proxy for landmark jitter without requiring explicit landmark estimation.

3.3.3 Temporal Flow Residual Features. Temporal incoherence characterizes manipulated videos, particularly when frames are generated independently. Dense optical flow between consecutive face crops is computed using the Farneback method (pyramid scale 0.5, 3 levels, window size 15, 3 iterations). Frame t is warped to approximate frame $t + 1$ via bilinear interpolation, and pixel-wise residual error between the warped and actual frames is measured. For each video, mean, standard deviation, maximum, and 95th percentile of residual magnitudes across consecutive frame pairs are aggregated into a 4-dimensional temporal flow residual feature.

3.3.4 Face Embeddings and Complementary Features. Face embeddings extracted from a pre-trained InceptionResnetV1 network (VGGFace2) serve as a baseline. Each cropped face is processed through the embedding model, and temporal descriptors are computed by tracking frame-to-frame changes in embedding space. Statistics include mean displacement, variance, kurtosis, lag-1 autocorrelation, and spectral centroid. When corresponding reference videos (source A and B) are available, differences between manipulated and authentic temporal descriptors emphasize identity and texture discrepancies. These embeddings primarily capture identity-level and texture-level cues, proving particularly informative for identity-swap methods.

Additionally, background embeddings are extracted by masking the face region and processing the remaining background through the same network. Pixel-level temporal features are computed by measuring frame-to-frame intensity differences and their high-frequency spectral energy.

3.3.5 Feature Fusion. All feature sets undergo z-score normalization before classification, with scaling parameters estimated exclusively on training folds to prevent information leakage. Experiments evaluate both individual feature families and concatenated representations. The most effective combination merges face embeddings (temporal descriptors)

with FFT features (6 dimensions), exploiting complementary spatial-frequency information.

3.4 Classification Models and Training Protocol

Three classification models are evaluated:

Random Forest (RF): The primary classifier, configured with 150 trees for one-vs-all and pairwise tasks and 300 trees for multi-class evaluation. Unlimited depth, class-balanced weights, and Gini impurity criterion are employed.

Support Vector Machine (SVM): Utilizes an RBF kernel with regularization parameter $C = 1.0$ and automatic gamma scaling.

XGBoost (XGB): Configured with 150 trees, maximum depth of 6, learning rate of 0.05, subsample ratio of 0.8, column subsample ratio of 0.8, and histogram-based tree construction.

Training and evaluation employ stratified 10-fold cross-validation (shuffle enabled, random state 0) to preserve class balance. Feature extraction, normalization, and model fitting are performed independently within each fold. Three prediction tasks are evaluated:

- (1) **One-vs-all classification:** Each method is classified against all others, yielding four binary problems evaluated using ROC-AUC.
- (2) **Pairwise binary classification:** All six method pairs are evaluated for discrimination ability using accuracy and ROC-AUC metrics.
- (3) **Four-way multi-class classification:** Direct prediction of manipulation method among all four classes, evaluated using accuracy, one-vs-rest ROC-AUC, and weighted one-vs-rest ROC-AUC. Confusion matrices are generated for error analysis.

For feature combination experiments (face embeddings + FFT), StandardScaler is applied within a pipeline before the Random Forest classifier, and pairwise AUC performance is compared across individual and combined feature sets.

4 EXPERIMENTS

4.1 Experimental Setup

We evaluate our approach on the FaceForensics++ dataset (c23 compression), which contains 480 videos equally distributed across four manipulation methods: DeepFakes, Face2Face, FaceSwap, and NeuralTextures. From each video, we extract 20-30 uniformly sampled frames and perform face detection using MTCNN [25]. We exclude videos with fewer than 18 detected faces to ensure reliable temporal statistics.

Table 1: Multi-class classification performance (4-way) with Random Forest. Random baseline: 25% accuracy, 50% AUC. Best results in bold.

Feature Type	Accuracy (%)	AUC (%)
FFT (Radial)	26.0 ± 6.3	54.3 ± 4.3
Landmark Jitter	24.6 ± 5.7	47.6 ± 6.6
Optical Flow	25.6 ± 5.7	49.7 ± 4.8
Face Embeddings	38.8 ± 6.2	66.9 ± 4.5
Background	20.8 ± 4.6	39.9 ± 3.1
Pixel Temporal	24.0 ± 4.9	47.8 ± 5.3
Random Baseline	25.0	50.0

4.2 Evaluation Protocol

We conduct two complementary evaluations: pairwise classification across all six method pairs and four-way multi-class classification. Both tasks employ stratified 10-fold cross-validation [13] to preserve class balance. To prevent data leakage, we perform all feature extraction and normalization within each fold. We measure performance using accuracy and area under the ROC curve (AUC), computing binary AUC for pairwise tasks and one-vs-rest macro AUC for multi-class classification.

4.3 Classification Models

We employ Random Forest [4] as our primary classifier due to its robustness to noisy features and ability to model non-linear decision boundaries. Our model uses 150 trees with unlimited depth and balanced class weights. We also evaluate Support Vector Machines (RBF kernel, $C = 1.0$) and XGBoost (150 trees, depth=6, learning rate=0.05) for comparison, using identical cross-validation splits to ensure fairness.

Table 1 presents four-way classification performance across feature types. Face embeddings achieve the highest performance (38.8% accuracy, 66.9% AUC), substantially exceeding the 25% random baseline. FFT features demonstrate modest discrimination capability (26.0% accuracy, 54.3% AUC). In contrast, geometric and temporal features—landmark jitter and optical flow—perform near or below chance (47.6% and 49.7% AUC, respectively). Background features achieve the lowest performance (20.8% accuracy), confirming that manipulation artifacts concentrate in facial regions.

Figure 1 shows the normalized confusion matrix for FFT features, revealing systematic misclassification patterns aligned with technical similarity. DeepFakes videos are frequently misclassified as NeuralTextures (38%), consistent with both methods employing neural synthesis. Similarly, Face2Face and FaceSwap exhibit high mutual confusion (39% in each

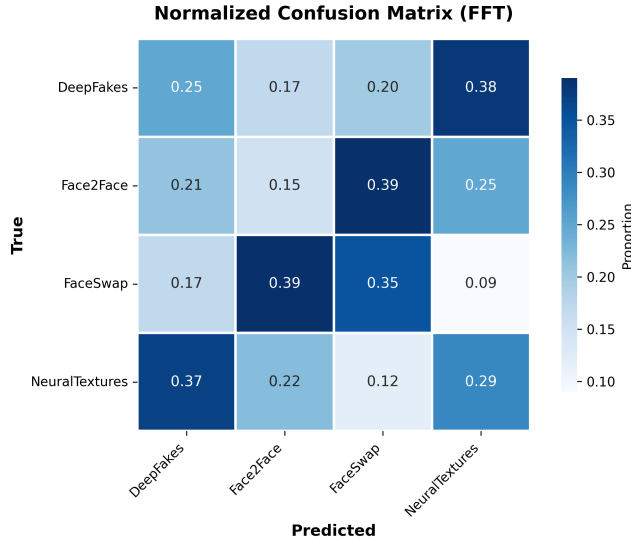


Figure 1: Normalized confusion matrix for FFT features showing systematic misclassifications between technically similar methods: DeepFakes↔NeuralTextures (neural synthesis) and Face2Face↔FaceSwap (3D-based manipulation).

direction), reflecting their shared reliance on 3D geometry-based manipulation. These patterns demonstrate a fundamental challenge: methods sharing underlying technical paradigms produce overlapping artifact distributions that single feature types cannot reliably separate.

4.4 Pairwise Discrimination

Figure 2 presents pairwise classification performance for FFT features, face embeddings, and their combination across all six method pairs. Performance varies dramatically by pair, with AUC ranging from 42 (FFT for DeepFakes vs. NeuralTextures) to 91% (face embeddings for FaceSwap vs. NeuralTextures). High Separability Pairs: Face embeddings achieve strong discrimination for FaceSwap vs. NeuralTextures (91 AUC) and DeepFakes vs. FaceSwap (85 AUC). Face2Face vs. FaceSwap also exhibits high separability (89% AUC), demonstrating that expression reenactment and identity replacement introduce fundamentally different artifacts detectable through identity-preserving features. Challenging Pairs: DeepFakes vs. Face2Face represents the most difficult discrimination task, with all features performing near random (FFT: 55 AUC; Face: 53% AUC). DeepFakes vs. NeuralTextures also proves challenging (56% AUC), as both methods employ neural synthesis producing similar spectral artifacts. Feature Specificity: FFT features prove most effective for pairs involving neural rendering methods, achieving 75 AUC

for FaceSwap vs. NeuralTextures and 69% AUC for DeepFakes vs. FaceSwap. However, FFT performance degrades when both methods use neural synthesis (42% AUC for DeepFakes vs. NeuralTextures).

4.5 Feature Combination

Concatenating face embeddings (512 dimensions) with FFT features (6 dimensions) yields modest improvements. As shown in Figure 2, the combined representation achieves 3-4 AUC gains for challenging pairs where neither feature alone excels (DeepFakes vs. Face2Face, DeepFakes vs. NeuralTextures). However, for pairs where face embeddings already dominate (FaceSwap vs. NeuralTextures, Face2Face vs. FaceSwap), combination provides negligible benefit (<1% AUC), indicating that face embeddings already capture the most discriminative information. These results suggest that feature combination helps primarily when individual features fail, offering limited advantage when strong features already exist.

5 CONCLUSIONS

This project demonstrates that deepfake method attribution is intrinsically heterogeneous. Different manipulation pipelines introduce different artifacts, and no single representation is sufficient. Key takeaways:

- Frequency-domain cues are critical for neural rendering methods
- Identity embeddings dominate classical swap methods
- Temporal instability provides complementary information
- Feature fusion must be applied selectively

Rather than optimizing for raw accuracy, this work emphasizes diagnostic understanding, offering insights into why some deepfakes are easier to distinguish than others.

6 NOTEBOOK

The notebook can be found here.

REFERENCES

- [1] Irene Amerini, Leonardo Galteri, Roberto Caldelli, and Alberto Del Bimbo. 2019. Deepfake Video Detection through Optical Flow Based CNN. In *Proceedings of the IEEE/CVF International Conference on Computer Vision Workshops (ICCVW)*. 1205–1207.
- [2] Muhammad Ahmad Amin, Yongjian Hu, and Jiankun Hu. 2024. Analyzing Temporal Coherence for Deepfake Video Detection. *Electronic Research Archive* 32, 4 (2024), 2621–2641.
- [3] Spiros Baxevanakis, Manos Schinas, and Symeon Papadopoulos. 2025. Do DeepFake Attribution Models Generalize?. In *Proceedings of the 4th ACM International Workshop on Multimedia AI against Disinformation*. 45–54.
- [4] Leo Breiman. 2001. Random Forests. *Machine Learning* 45, 1 (2001), 5–32.

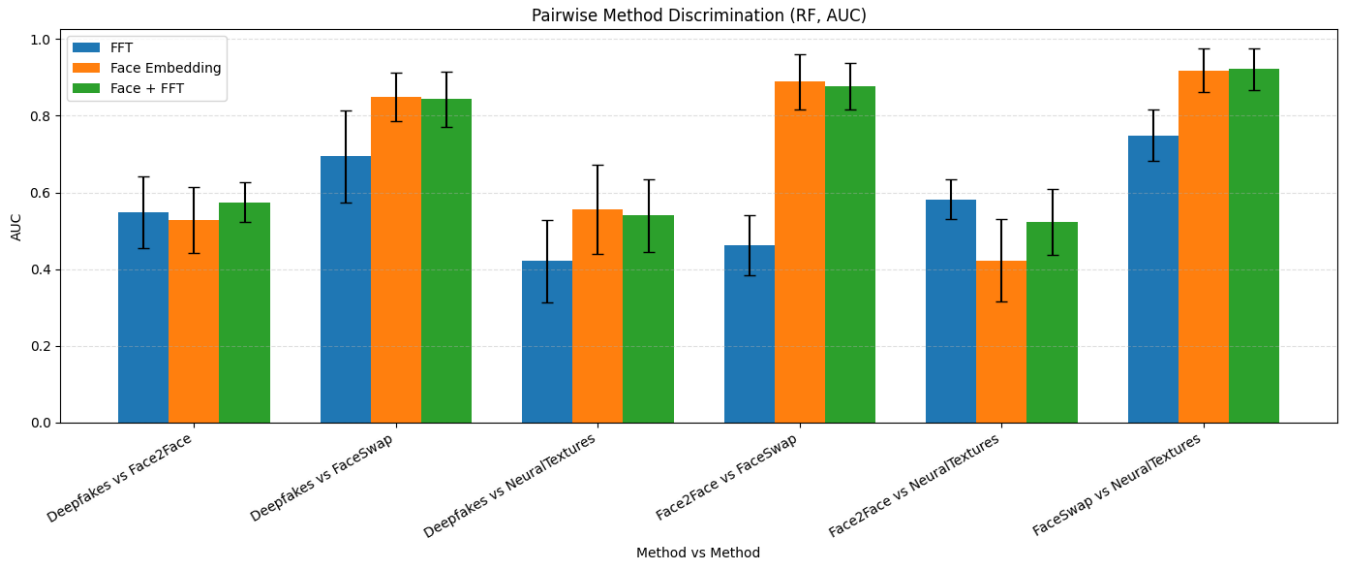


Figure 2: Pairwise classification performance (AUC) comparing FFT, face embeddings, and their combination. Error bars indicate standard deviation across 10 folds. No single feature dominates; effective discrimination requires matching features to manipulation mechanisms.

- [5] Roberto Caldelli, Leonardo Galteri, Irene Amerini, and Alberto Del Bimbo. 2021. Optical Flow Based CNN for Detection of Unlearned Deepfake Manipulations. *Pattern Recognition Letters* 146 (2021), 31–37.
- [6] François Chollet. 2017. Xception: Deep Learning with Depthwise Separable Convolutions. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*. 1251–1258.
- [7] Brian Dolhansky, Joanna Bitton, Ben Pfau, Jikuo Lu, Russ Howes, Menglin Wang, and Cristian Canton Ferrer. 2020. The DeepFake Detection Challenge (DFDC) Dataset. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*. 2065–2074.
- [8] Ricard Durall, Margret Keuper, Franz-Josef Pfundt, and Janis Keuper. 2020. Watch Your Up-Convolution: CNN Based Generative Deep Neural Networks Are Failing to Reproduce Spectral Distributions. *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)* (2020), 7890–7899.
- [9] Joel Frank, Thorsten Eisenhofer, Lea Schönherr, Asja Fischer, Dorothea Kolossa, and Thorsten Holz. 2020. Leveraging Frequency Analysis for Deep Fake Image Recognition. In *Proceedings of the 37th International Conference on Machine Learning (ICML)*. 3247–3258.
- [10] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. 2016. Deep Residual Learning for Image Recognition. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*. 770–778.
- [11] Asma Heidari, Nima Jafari Navimipour, Mehmet Unal, and Shiva Toulmaji. 2024. Deepfake Detection Using Deep Learning Methods: A Systematic and Comprehensive Review. *WIREs Data Mining and Knowledge Discovery* 14, 2 (2024), e1520.
- [12] Liming Jiang, Ren Li, Wayne Wu, Chen Qian, and Chen Change Loy. 2020. DeeperForensics-1.0: A Large-Scale Dataset for Real-World Face Forgery Detection. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 2886–2895.
- [13] Ron Kohavi. 1995. A Study of Cross-Validation and Bootstrap for Accuracy Estimation and Model Selection. 14 (1995), 1137–1145.
- [14] Yuezun Li and Siwei Lyu. 2019. Exposing Deepfake Videos By Detecting Face Warping Artifacts. In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR) Workshops*.
- [15] Yuezun Li, Xin Yang, Pu Sun, Honggang Qi, and Siwei Lyu. 2020. Celeb-DF: A Large-scale Challenging Dataset for DeepFake Forensics. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 3207–3216.
- [16] Yuchen Luo, Yong Zhang, Junchi Yan, and Wei Liu. 2021. Generalizing Face Forgery Detection with High-frequency Features. *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)* (2021), 16317–16326.
- [17] Falko Matern, Christian Riess, and Marc Stamminger. 2019. Exploiting Visual Artifacts to Expose Deepfakes and Face Manipulations. In *Proceedings of the IEEE Winter Applications of Computer Vision Workshops (WACVW)*. 83–92.
- [18] Yuyang Qian, Guojun Yin, Lu Sheng, Zixuan Chen, and Jing Shao. 2020. Thinking in Frequency: Face Forgery Detection by Mining Frequency-aware Clues. In *Proceedings of the European Conference on Computer Vision (ECCV)*. 86–103.
- [19] Andreas Rössler, Davide Cozzolino, Luisa Verdoliva, Christian Riess, Justus Thies, and Matthias Nießner. 2019. FaceForensics++: Learning to Detect Manipulated Facial Images. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*. 1–11.
- [20] Chuangchuang Tan, Yao Zhao, Shikui Wei, Guanghua Gu, Ping Liu, and Yunchao Wei. 2024. Frequency-Aware Deepfake Detection: Improving Generalizability through Frequency Space Learning. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 38. 5178–5186.
- [21] Justus Thies, Michael Zollhöfer, and Matthias Nießner. 2019. Deferred Neural Rendering: Image Synthesis Using Neural Textures. *ACM Transactions on Graphics (TOG)* 38, 4 (2019), 1–12.
- [22] Justus Thies, Michael Zollhöfer, Marc Stamminger, Christian Theobalt, and Matthias Nießner. 2016. Face2Face: Real-time Face Capture and

- Reenactment of RGB Videos. *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)* (2016), 2387–2395.
- [23] Xiaoyi Wang, Yihao Huang, Chuangchuang Tan, Yao Zhao, and Yun-chao Wei. 2023. AltFreezing: Alternately Freezing Spatial and Temporal Networks for Efficient Deepfake Video Detection. *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)* (2023), 13863–13872.
- [24] Ning Yu, Larry S Davis, and Mario Fritz. 2019. Attributing Fake Images to GANs: Learning and Analyzing GAN Fingerprints. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*. 7556–7566.
- [25] Kaipeng Zhang, Zhanpeng Zhang, Zhifeng Li, and Yu Qiao. 2016. Joint Face Detection and Alignment Using Multitask Cascaded Convolutional Networks. *IEEE Signal Processing Letters* 23, 10, 1499–1503.
- [26] Yinglin Zheng, Jianmin Bao, Dong Chen, Ming Zeng, and Fang Wen. 2021. Exploring Temporal Coherence for More General Video Face Forgery Detection. *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)* (2021), 15044–15054.